

GPU scaling runs: ALIGNN-FF and VASP

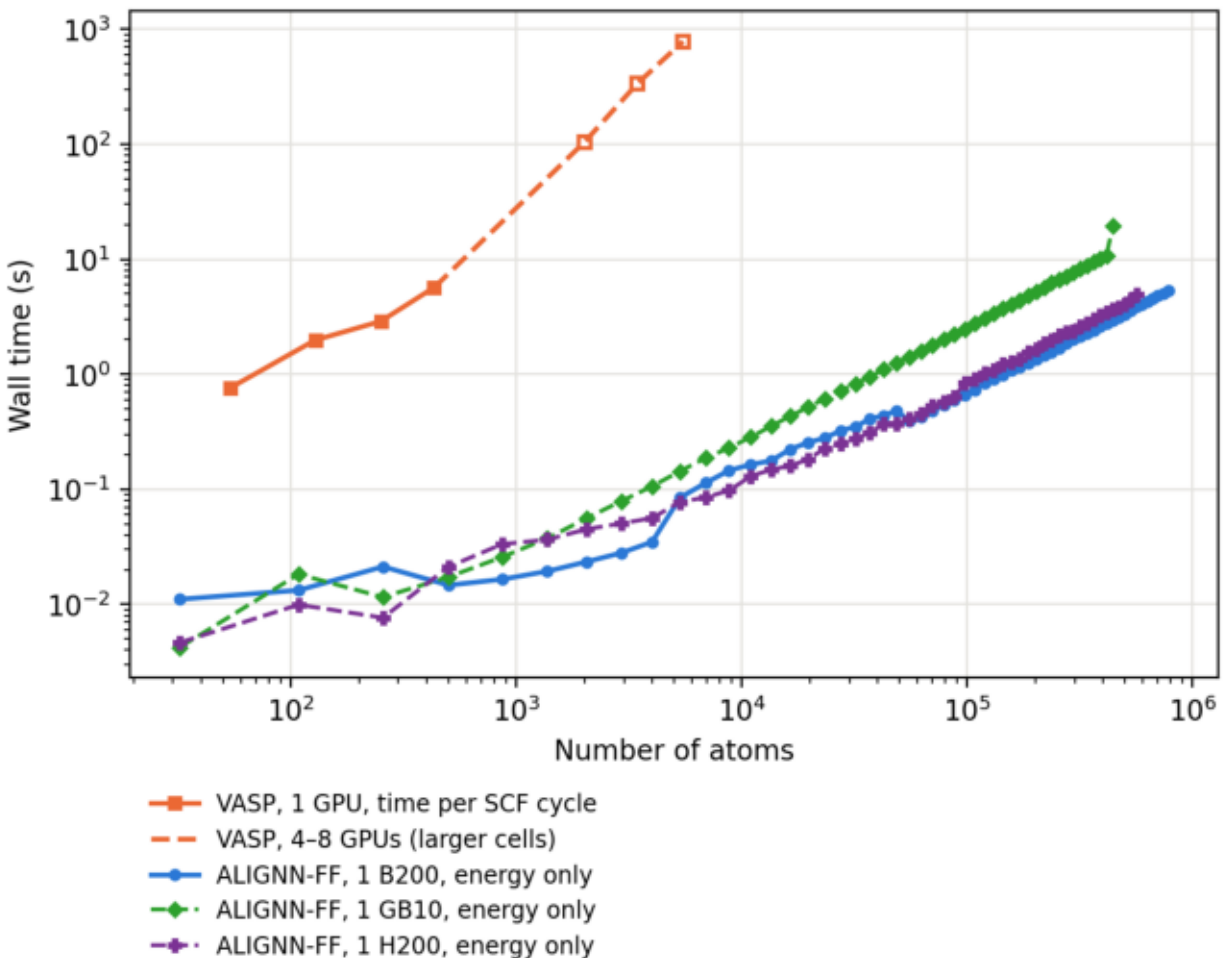
B200 runs: NVIDIA B200 GPUs (SLURM partition b200, QOS blackwell_test).

ALIGNN-FF, energy only (no forces), Cu FCC, 1 B200: 58 supercells, $1\times 1\times 1$ to $58\times 58\times 58$ (4 to 780,448 atoms). Energies correct up to 442,368 atoms ($48\times 48\times 48$); larger sizes are affected by the float32 drift and are shaded red in the table.

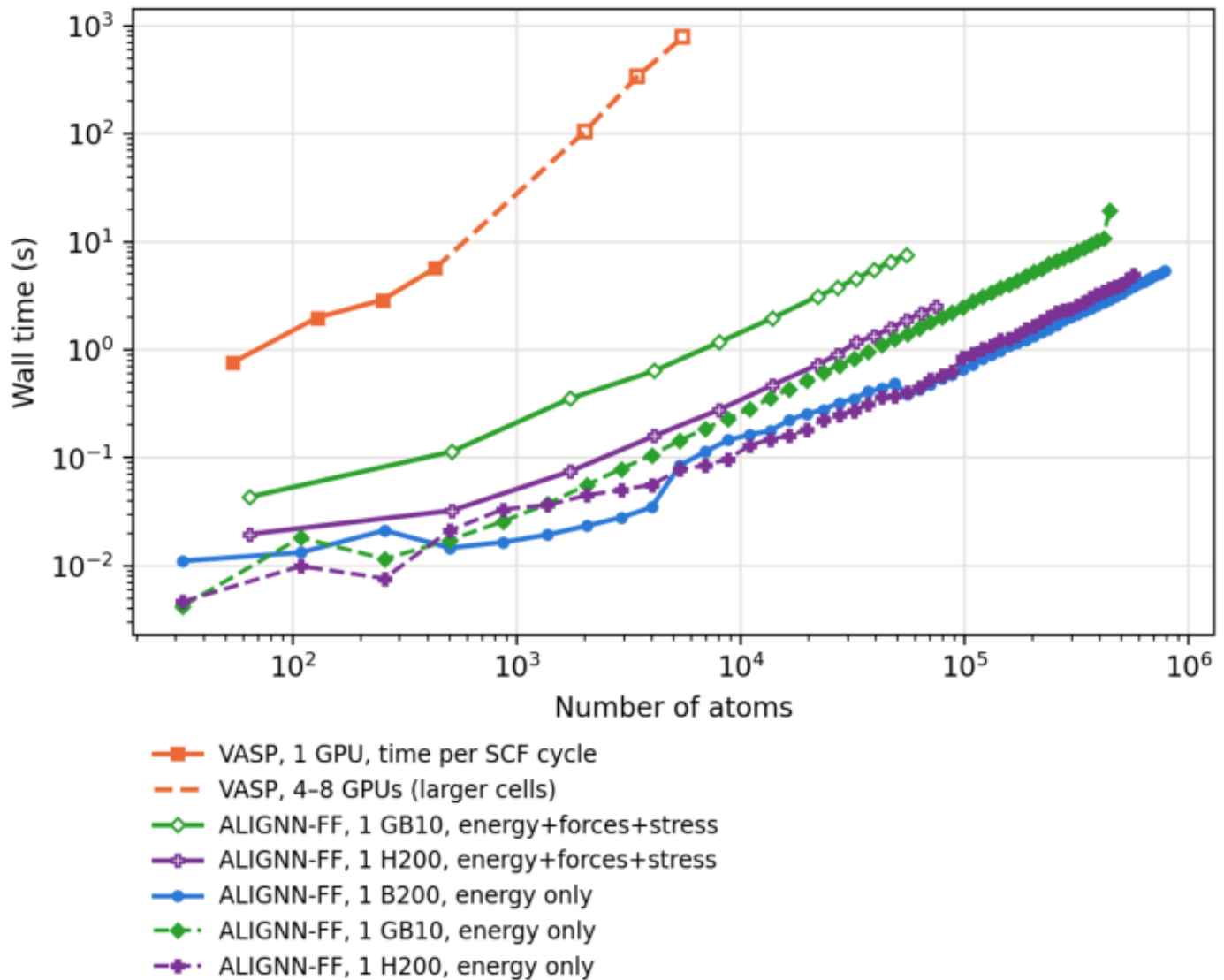
VASP, Si diamond: 21 runs over 9 supercell sizes (54 to 8,192 atoms); 19 completed, 2 crashed before the first SCF cycle ($15\times 15\times 15$ and $16\times 16\times 16$).

ALIGNN-FF, energy only, B200 settings rerun on a GB10 and an H200 (Cu FCC, same checkpoint, pure-PyTorch model; energies match the B200 to $4e-6$ eV/atom); largest cell 442,368 atoms (GB10) and 562,432 (H200). At 256,000 atoms, inference takes 5.48 s (GB10), 0.70 s (H200) and 0.62 s (B200).

ALIGNN-FF with energy + forces + stress (added 2026-09-27; the MD workload), Si diamond, 1 GPU, default matpes_r2scan checkpoint (hidden 128, smooth cutoff, 52 neighbours), same script on both cards: GB10 (121.7 GiB): largest cell 54,872 atoms at 107 GB, OOM at 64,000; H200 NVL (139.8 GiB): 74,088 atoms at 145 GB, OOM at 85,184. Memory 1.95 MB/atom on both; the H200 is 4× faster than the GB10 at every size.



All runs: VASP and ALIGNN-FF on B200, GB10 and H200



Energy only: filled markers (B200 run and its GB10/H200 reruns, Cu, 12 neighbours). Energy + forces + stress: hollow markers, solid lines (Si, smooth 52-neighbour matpes_r2scan, the MD workload). VASP: time per SCF cycle, Si.

Which GPU for which job

	GB10	H200 NVL	B200
Energy only, 256,000 atoms: inference (s)	5.48	0.70	0.62
Energy only, 256,000 atoms: graph build (s, CPU)	1.12	1.50	1.07
Energy only: largest cell (atoms)	442,368	562,432	780,448
Energy + forces + stress, 21,952 atoms (s)	3.12	0.72	not run
Energy + forces + stress: largest cell (atoms)	54,872	74,088	not run
GPU memory	121.7 GiB (unified)	139.8 GiB	178.3 GiB

Reading the numbers

- Model evaluation: the H200 is within 13% of the B200 and 8x faster than the GB10 (energy only, identical settings and checkpoint). With forces and stress the H200 is 4.3x the GB10.
- Graph construction runs on the host CPU, so it follows the machine, not the GPU.
- The largest cell is set by memory alone: 1.95 MB per atom with forces, 0.25 MB/atom energy only, the same on the GB10 and H200.

Recommendation

- Long training runs: GB10. Slowest per GPU, but locally available at no cost; multi-node training, which would recover much of the per-GPU gap, is still being made to work.
- Large-cell MD and fast turnaround: H200. Largest cell with forces we can reach (74k atoms) and ~4x the GB10, but it is a shared, allocation-billed resource.
- B200: fastest and largest on paper, but not available to this project; the H200 comes within 13% of it on model evaluation, so it is not worth pursuing.

Caveats: single-point evaluations, not full MD or training steps, so the ratios will shift somewhat for real workloads; multi-node GB10 training throughput is not yet measured.

Scaling to one billion atoms (with forces)

GPU	Atoms per GPU	GPUs, no halo	Halo factor	GPUs with halo
GB10	54,872 (measured)	18,224	2.7x	48,702
H200 NVL	74,088 (measured)	13,497	2.5x	33,267
MI250X GCD (Frontier)	33,917 (memory rule)	29,484	3.1x	90,845 (= 11,356 nodes)

The governing number

- With forces the force field needs 1.95 MB of GPU memory per atom on every card measured, so 10^9 atoms is ~ 2.0 PB before overheads. 'Memory rule' rows apply that figure, and the 96% of memory the H200 could use, to the card's memory; they are estimates, not measurements.
- ALIGNN is a message-passing network, so when the crystal is split across GPUs each GPU also holds a ghost shell as thick as the receptive field (~ 20 Å here: $2+2$ layers $\times 5$ Å). Around a ~ 70 k-atom Si block that multiplies the atoms each GPU holds by the halo factor. As-is, the GCD count exceeds Frontier (9,408 nodes).

Levers, largest first

- Memory per atom: the energy-only model uses 0.25 MB/atom (hidden 64, 12 neighbours); the 52-neighbour line graph dominates. A leaner force field, half precision and dropping stress could cut 1.95 MB several-fold (unmeasured, and accuracy must be re-checked).
- Fewer message-passing layers shrink the halo: 20 Å $\rightarrow$ 10 Å takes the factor from ~ 2.5 x to ~ 1.6 x.
- Memory per GPU matters more than speed: the job is memory-bound.

Software

- A run this size needs LAMMPS domain decomposition with ghost-atom exchange. ALIGNN has a LAMMPS interface (pair_alignn, TorchScript); multi-layer ghost handling and an AMD build are not yet verified.
- The path is a ladder: 1 GPU $\rightarrow$ 1 node $\rightarrow$ 64 nodes $\rightarrow$ thousands, fixing scaling at each step.

Time, if it fits

- At ~ 2.5 s per step per GPU (H200 at its largest cell), 1 ps at 1 fs steps is ~ 42 min of wall time plus communication; at thousands of nodes, every hour costs thousands of node-hours.

First steps (cheap, one GB10): memory per atom with forces for a leaner model (fewer neighbours, fewer layers, half precision), and pair_alignn on 2 GPUs with a real ghost region.

ALIGNN-FF results, energy only, B200 (1/2)

Supercell	Atoms	Neighbor list (s)	Graph (s)	Inference (s)	Total (s)	Model output (eV/atom)
1×1×1	4	0.0844	0.1568	3.2238	3.4650	0.604015
2×2×2	32	0.0015	0.0022	0.0073	0.0111	0.604014
3×3×3	108	0.0018	0.0026	0.0089	0.0132	0.604014
4×4×4	256	0.0087	0.0026	0.0100	0.0212	0.604014
5×5×5	500	0.0035	0.0029	0.0083	0.0147	0.604013
6×6×6	864	0.0047	0.0031	0.0086	0.0165	0.604013
7×7×7	1,372	0.0063	0.0035	0.0096	0.0194	0.604013
8×8×8	2,048	0.0085	0.0040	0.0109	0.0234	0.604013
9×9×9	2,916	0.0115	0.0046	0.0118	0.0279	0.604013
10×10×10	4,000	0.0158	0.0055	0.0136	0.0348	0.604013
11×11×11	5,324	0.0433	0.0181	0.0231	0.0845	0.604013
12×12×12	6,912	0.0680	0.0151	0.0304	0.1135	0.604013
13×13×13	8,788	0.0733	0.0248	0.0476	0.1457	0.604013
14×14×14	10,976	0.0881	0.0216	0.0542	0.1639	0.604013
15×15×15	13,500	0.1020	0.0252	0.0498	0.1771	0.604013
16×16×16	16,384	0.1218	0.0323	0.0682	0.2223	0.604013
17×17×17	19,652	0.1479	0.0399	0.0669	0.2547	0.604013
18×18×18	23,328	0.1598	0.0403	0.0786	0.2787	0.604013
19×19×19	27,436	0.1912	0.0387	0.0928	0.3226	0.604013
20×20×20	32,000	0.2106	0.0414	0.0985	0.3506	0.604013
21×21×21	37,044	0.2482	0.0462	0.1152	0.4096	0.604013
22×22×22	42,592	0.2590	0.0495	0.1283	0.4368	0.604013
23×23×23	48,668	0.2798	0.0548	0.1499	0.4845	0.604013
24×24×24	55,296	0.2096	0.0429	0.1373	0.3897	0.604013
25×25×25	62,500	0.2265	0.0481	0.1548	0.4294	0.604013
26×26×26	70,304	0.2547	0.0537	0.1738	0.4822	0.604015
27×27×27	78,732	0.2841	0.0601	0.1935	0.5377	0.604013
28×28×28	87,808	0.3112	0.0651	0.2156	0.5919	0.604013
29×29×29	97,556	0.3462	0.0750	0.2390	0.6602	0.604013
30×30×30	108,000	0.3773	0.0811	0.2643	0.7227	0.604013

ALIGNN-FF v6 (scale5b_v6.py), one B200, float32. Times are wall time for one energy evaluation.
The first row includes one-time GPU warm-up. Red rows (N > 442,368): the model output is wrong because of the float32 drift; the float64 reference is 0.604015 eV/atom.

ALIGNN-FF results, energy only, B200 (2/2)

Supercell	Atoms	Neighbor list (s)	Graph (s)	Inference (s)	Total (s)	Model output (eV/atom)
31×31×31	119,164	0.4516	0.0884	0.2913	0.8313	0.604013
32×32×32	131,072	0.4864	0.0953	0.3193	0.9011	0.604013
33×33×33	143,748	0.5267	0.1030	0.3502	0.9799	0.604013
34×34×34	157,216	0.5839	0.1139	0.3820	1.0798	0.604013
35×35×35	171,500	0.6101	0.1259	0.4164	1.1525	0.604013
36×36×36	186,624	0.6554	0.1350	0.4529	1.2434	0.604013
37×37×37	202,612	0.7091	0.1466	0.4912	1.3469	0.604013
38×38×38	219,488	0.7572	0.1677	0.5319	1.4568	0.604013
39×39×39	237,276	0.8124	0.1740	0.5753	1.5617	0.604013
40×40×40	256,000	0.8806	0.1872	0.6202	1.6880	0.604013
41×41×41	275,684	1.0001	0.1986	0.6675	1.8662	0.604013
42×42×42	296,352	1.0695	0.2132	0.7176	2.0003	0.604013
43×43×43	318,028	1.1327	0.2275	0.7691	2.1293	0.604015
44×44×44	340,736	1.1961	0.2483	0.8239	2.2683	0.604013
45×45×45	364,500	1.2776	0.2635	0.8810	2.4221	0.604013
46×46×46	389,344	1.3576	0.2852	0.9407	2.5835	0.604015
47×47×47	415,292	1.4514	0.3117	1.0032	2.7663	0.604015
48×48×48	442,368	1.5614	0.3171	1.0683	2.9467	0.604013
49×49×49	470,596	1.6539	0.3416	1.1366	3.1321	0.606979
50×50×50	500,000	1.7436	0.3695	1.2075	3.3206	0.639071
51×51×51	530,604	1.9523	0.3813	1.2820	3.6156	0.672090
52×52×52	562,432	2.0831	0.4097	1.3590	3.8518	0.702621
53×53×53	595,508	2.1964	0.4336	1.4393	4.0692	0.730894
54×54×54	629,856	2.3269	0.4530	1.5224	4.3024	0.757119
55×55×55	665,500	2.4602	0.4827	1.6091	4.5520	0.781473
56×56×56	702,464	2.5851	0.5138	1.6982	4.7971	0.804124
57×57×57	740,772	2.7414	0.5370	1.7917	5.0702	0.825215
58×58×58	780,448	2.9055	0.5709	1.9083	5.3847	0.844881

ALIGNN-FF v6 (scale5b_v6.py), one B200, float32. Times are wall time for one energy evaluation.
The first row includes one-time GPU warm-up. Red rows (N > 442,368): the model output is wrong because of the float32 drift; the float64 reference is 0.604015 eV/atom.

ALIGNN-FF results, energy only, GB10 (1/2)

Supercell	Atoms	Graph (s)	Inference (s)	Total (s)	Peak mem (GB)	Model output (eV/atom)
1×1×1	4	1.3637	0.0701	1.4338	0.0	0.604014
2×2×2	32	0.0017	0.0025	0.0042	0.0	0.604015
3×3×3	108	0.0145	0.0037	0.0182	0.1	0.604015
4×4×4	256	0.0050	0.0066	0.0115	0.1	0.604020
5×5×5	500	0.0053	0.0120	0.0173	0.2	0.604017
6×6×6	864	0.0068	0.0189	0.0257	0.3	0.604021
7×7×7	1,372	0.0087	0.0287	0.0374	0.4	0.604014
8×8×8	2,048	0.0122	0.0438	0.0560	0.6	0.604022
9×9×9	2,916	0.0148	0.0639	0.0787	0.8	0.604024
10×10×10	4,000	0.0195	0.0865	0.1060	1.0	0.604012
11×11×11	5,324	0.0279	0.1159	0.1437	1.4	0.604015
12×12×12	6,912	0.0326	0.1538	0.1864	1.8	0.604044
13×13×13	8,788	0.0367	0.1924	0.2291	2.3	0.604008
14×14×14	10,976	0.0458	0.2384	0.2841	2.8	0.603933
15×15×15	13,500	0.0608	0.2931	0.3539	3.5	0.603850
16×16×16	16,384	0.0770	0.3540	0.4310	4.2	0.603873
17×17×17	19,652	0.0910	0.4241	0.5151	5.0	0.603919
18×18×18	23,328	0.1104	0.5035	0.6139	5.9	0.603956
19×19×19	27,436	0.1265	0.5906	0.7171	7.0	0.603956
20×20×20	32,000	0.1427	0.6841	0.8268	8.1	0.603860
21×21×21	37,044	0.1612	0.7881	0.9493	9.4	0.603743
22×22×22	42,592	0.1921	0.9065	1.0986	10.8	0.603745
23×23×23	48,668	0.2169	1.0358	1.2527	12.4	0.603927
24×24×24	55,296	0.2349	1.1735	1.4083	14.0	0.604198
25×25×25	62,500	0.2678	1.3306	1.5985	15.9	0.604642
26×26×26	70,304	0.3060	1.4957	1.8016	17.8	0.605064
27×27×27	78,732	0.3374	1.6727	2.0101	20.0	0.605457
28×28×28	87,808	0.3540	1.8783	2.2323	22.3	0.605647
29×29×29	97,556	0.4018	2.0760	2.4778	24.8	0.605734
30×30×30	108,000	0.4980	2.2980	2.7959	27.4	0.605611

B200 settings rerun on one NVIDIA GB10: Cu FCC, 5 Å cutoff, 12 neighbours, energy only, float32, checkpoint v12.2.2024_dft_3d_307k, loaded into the pure-PyTorch ALIGNN (reproduces the B200 DGL energies to 4e-6 eV/atom). One call per size; the first row includes GPU warm-up. Largest size shown is the largest that completed; the next (49×49×49 = 470,596 atoms) did not fit. Red rows (N > 442,368): float32 energy drift, as on the B200. The port has no float64 readout, so smaller cells also drift slightly (to ~0.600). Timings are unaffected. atomgptlab job 13484, 2026-09-28.

ALIGNN-FF results, energy only, GB10 (2/2)

Supercell	Atoms	Graph (s)	Inference (s)	Total (s)	Peak mem (GB)	Model output (eV/atom)
31×31×31	119,164	0.5406	2.5462	3.0868	30.2	0.604993
32×32×32	131,072	0.5858	2.7901	3.3759	33.2	0.604449
33×33×33	143,748	0.6294	3.1341	3.7635	36.5	0.603988
34×34×34	157,216	0.6864	3.3575	4.0439	39.9	0.603673
35×35×35	171,500	0.7332	3.6670	4.4002	43.5	0.602742
36×36×36	186,624	0.8208	3.9880	4.8089	47.3	0.602057
37×37×37	202,612	0.8541	4.4270	5.2811	51.4	0.601320
38×38×38	219,488	0.9349	4.6847	5.6196	55.6	0.601225
39×39×39	237,276	1.0287	5.1840	6.2126	60.1	0.600929
40×40×40	256,000	1.1166	5.4838	6.6004	64.9	0.600853
41×41×41	275,684	1.1886	5.8977	7.0864	69.9	0.600795
42×42×42	296,352	1.2488	6.3326	7.5814	75.1	0.600620
43×43×43	318,028	1.3689	6.8675	8.2364	80.6	0.600555
44×44×44	340,736	1.4533	7.3131	8.7664	86.4	0.600509
45×45×45	364,500	1.4925	8.0306	9.5232	92.4	0.600274
46×46×46	389,344	1.7065	8.4010	10.1075	98.7	0.600081
47×47×47	415,292	1.8319	8.9564	10.7883	105.2	0.599898
48×48×48	442,368	1.9177	17.4482	19.3659	112.1	0.600148

B200 settings rerun on one NVIDIA GB10: Cu FCC, 5 Å cutoff, 12 neighbours, energy only, float32, checkpoint v12.2.2024_dft_3d_307k, loaded into the pure-PyTorch ALIGNN (reproduces the B200 DGL energies to 4e-6 eV/atom). One call per size; the first row includes GPU warm-up. Largest size shown is the largest that completed; the next (49×49×49 = 470,596 atoms) did not fit. Red rows (N > 442,368): float32 energy drift, as on the B200. The port has no float64 readout, so smaller cells also drift slightly (to ~0.600). Timings are unaffected. atomgptlab job 13484, 2026-09-28.

ALIGNN-FF results with forces, GB10

Supercell	Atoms	Time (s)	Peak mem (GB)	MB/atom	Energy (eV/atom)	Max F (eV/A)
2×2×2	64	0.043	0.2	3.03	-8.7476	2.0e-05
4×4×4	512	0.114	1.1	2.09	-8.7476	3.5e-05
6×6×6	1,728	0.353	3.4	1.99	-8.7476	3.8e-05
8×8×8	4,096	0.635	8.1	1.97	-8.7475	3.8e-05
10×10×10	8,000	1.174	15.7	1.96	-8.7479	3.9e-05
12×12×12	13,824	1.959	27.0	1.96	-8.7479	3.9e-05
14×14×14	21,952	3.116	42.9	1.95	-8.7476	5.6e-05
15×15×15	27,000	3.742	52.7	1.95	-8.7470	5.7e-05
16×16×16	32,768	4.576	64.0	1.95	-8.7464	5.8e-05
17×17×17	39,304	5.504	76.8	1.95	-8.7467	5.7e-05
18×18×18	46,656	6.454	91.1	1.95	-8.7470	5.7e-05
19×19×19	54,872	7.592	107.1	1.95	-8.7475	5.7e-05
20×20×20	64,000	OOM	-	-	-	-

Energy + forces + stress single point on one NVIDIA GB10: Si diamond (8-atom conventional cell, $i \times i \times i$), default matpes_r2scan checkpoint (hidden 128, 2+2 layers, smooth 5 Å cutoff, 52 neighbours), ASE calculator, float32. Time = median of 3 calls. Peak mem = peak allocated GPU memory. Max |F| should be ~ 0 for the perfect crystal (checks the forces). Red row: the first size that ran out of GPU memory. atomgptlab job 13474, 2026-09-27.

ALIGNN-FF results, energy only, H200 (1/2)

Supercell	Atoms	Graph (s)	Inference (s)	Total (s)	Peak mem (GB)	Model output (eV/atom)
1×1×1	4	0.5613	0.0668	0.6281	0.0	0.604015
2×2×2	32	0.0024	0.0023	0.0046	0.0	0.604017
3×3×3	108	0.0074	0.0025	0.0099	0.1	0.604017
4×4×4	256	0.0051	0.0025	0.0076	0.1	0.604020
5×5×5	500	0.0180	0.0032	0.0211	0.2	0.604019
6×6×6	864	0.0298	0.0035	0.0333	0.3	0.604021
7×7×7	1,372	0.0316	0.0049	0.0365	0.4	0.604014
8×8×8	2,048	0.0382	0.0067	0.0449	0.6	0.604023
9×9×9	2,916	0.0411	0.0093	0.0504	0.8	0.604024
10×10×10	4,000	0.0441	0.0121	0.0562	1.0	0.604011
11×11×11	5,324	0.0619	0.0157	0.0776	1.4	0.604010
12×12×12	6,912	0.0645	0.0203	0.0848	1.8	0.604036
13×13×13	8,788	0.0721	0.0254	0.0976	2.3	0.604000
14×14×14	10,976	0.0987	0.0313	0.1301	2.8	0.603929
15×15×15	13,500	0.1098	0.0382	0.1480	3.5	0.603846
16×16×16	16,384	0.1138	0.0460	0.1598	4.2	0.603869
17×17×17	19,652	0.1256	0.0561	0.1818	5.0	0.603917
18×18×18	23,328	0.1608	0.0648	0.2257	5.9	0.603956
19×19×19	27,436	0.1737	0.0762	0.2499	7.0	0.603955
20×20×20	32,000	0.1851	0.0884	0.2735	8.1	0.603858
21×21×21	37,044	0.2089	0.1022	0.3110	9.4	0.603743
22×22×22	42,592	0.2497	0.1174	0.3671	10.8	0.603744
23×23×23	48,668	0.2379	0.1339	0.3718	12.4	0.603927
24×24×24	55,296	0.2551	0.1519	0.4070	14.0	0.604197
25×25×25	62,500	0.2764	0.1720	0.4484	15.9	0.604641
26×26×26	70,304	0.3377	0.1929	0.5306	17.8	0.605063
27×27×27	78,732	0.3614	0.2159	0.5773	20.0	0.605456
28×28×28	87,808	0.3882	0.2408	0.6291	22.3	0.605646
29×29×29	97,556	0.5792	0.2674	0.8466	24.8	0.605734
30×30×30	108,000	0.6158	0.2960	0.9118	27.4	0.605611

B200 settings rerun on one NVIDIA H200 NVL: Cu FCC, 5 Å cutoff, 12 neighbours, energy only, float32, checkpoint v12.2.2024_dft_3d_307k, loaded into the pure-PyTorch ALIGNN (reproduces the B200 DGL energies to 4e-6 eV/atom). One call per size; the first row includes GPU warm-up. Largest size shown is the largest that completed; the next (53×53×53 = 595,508 atoms) did not fit. Red rows (N > 442,368): float32 energy drift, as on the B200. The port has no float64 readout, so smaller cells also drift slightly (to ~0.600). Timings are unaffected. Skipjack job 920089, 2026-09-28.

ALIGNN-FF results, energy only, H200 (2/2)

Supercell	Atoms	Graph (s)	Inference (s)	Total (s)	Peak mem (GB)	Model output (eV/atom)
31×31×31	119,164	0.6895	0.3260	1.0156	30.2	0.604993
32×32×32	131,072	0.7279	0.3587	1.0866	33.2	0.604449
33×33×33	143,748	0.8287	0.3946	1.2233	36.5	0.603987
34×34×34	157,216	0.8357	0.4293	1.2650	39.9	0.603673
35×35×35	171,500	0.8934	0.4685	1.3619	43.5	0.602742
36×36×36	186,624	1.0349	0.5095	1.5444	47.3	0.602057
37×37×37	202,612	1.1004	0.5565	1.6569	51.4	0.601320
38×38×38	219,488	1.2563	0.5999	1.8562	55.6	0.601225
39×39×39	237,276	1.3330	0.6514	1.9845	60.1	0.600928
40×40×40	256,000	1.5010	0.7002	2.2011	64.9	0.600853
41×41×41	275,684	1.5399	0.7535	2.2934	69.9	0.600795
42×42×42	296,352	1.5578	0.8105	2.3683	75.1	0.600620
43×43×43	318,028	1.7216	0.8683	2.5899	80.6	0.600554
44×44×44	340,736	1.8298	0.9311	2.7609	86.4	0.600510
45×45×45	364,500	2.0288	1.0003	3.0292	92.4	0.600274
46×46×46	389,344	2.1526	1.0680	3.2206	98.7	0.600081
47×47×47	415,292	2.3286	1.1412	3.4699	105.3	0.599898
48×48×48	442,368	2.4385	1.2090	3.6476	112.1	0.600148
49×49×49	470,596	2.4171	1.4137	3.8308	119.3	0.602111
50×50×50	500,000	2.5177	1.5239	4.0417	126.7	0.619691
51×51×51	530,604	2.7645	1.6988	4.4633	134.5	0.639844
52×52×52	562,432	2.9893	1.9614	4.9507	142.5	0.658485

B200 settings rerun on one NVIDIA H200 NVL: Cu FCC, 5 Å cutoff, 12 neighbours, energy only, float32, checkpoint v12.2.2024_dft_3d_307k, loaded into the pure-PyTorch ALIGNN (reproduces the B200 DGL energies to 4e-6 eV/atom). One call per size; the first row includes GPU warm-up. Largest size shown is the largest that completed; the next (53×53×53 = 595,508 atoms) did not fit. Red rows (N > 442,368): float32 energy drift, as on the B200. The port has no float64 readout, so smaller cells also drift slightly (to ~0.600). Timings are unaffected. Skipjack job 920089, 2026-09-28.

ALIGNN-FF results with forces, H200

Supercell	Atoms	Time (s)	Peak mem (GB)	MB/atom	Energy (eV/atom)	Max F (eV/A)
2×2×2	64	0.020	0.2	3.05	-8.7476	2.2e-05
4×4×4	512	0.032	1.1	2.09	-8.7476	3.6e-05
6×6×6	1,728	0.075	3.4	1.99	-8.7476	3.7e-05
8×8×8	4,096	0.159	8.1	1.97	-8.7475	3.9e-05
10×10×10	8,000	0.278	15.7	1.96	-8.7479	3.8e-05
12×12×12	13,824	0.470	27.0	1.96	-8.7479	4.0e-05
14×14×14	21,952	0.724	42.9	1.95	-8.7476	5.6e-05
15×15×15	27,000	0.905	52.7	1.95	-8.7470	5.8e-05
16×16×16	32,768	1.175	64.0	1.95	-8.7464	5.6e-05
17×17×17	39,304	1.339	76.8	1.95	-8.7467	5.7e-05
18×18×18	46,656	1.578	91.1	1.95	-8.7470	5.7e-05
19×19×19	54,872	1.887	107.1	1.95	-8.7475	5.6e-05
20×20×20	64,000	2.184	124.9	1.95	-8.7482	5.8e-05
21×21×21	74,088	2.534	144.6	1.95	-8.7482	5.8e-05
22×22×22	85,184	OOM	-	-	-	-

Energy + forces + stress single point on one NVIDIA H200 NVL: Si diamond (8-atom conventional cell, $i \times i \times i$), default matpes_r2scan checkpoint (hidden 128, 2+2 layers, smooth 5 Å cutoff, 52 neighbours), ASE calculator, float32. Time = median of 3 calls. Peak mem = peak allocated GPU memory. Max |F| should be ~0 for the perfect crystal (checks the forces). Red row: the first size that ran out of GPU memory. Skipjack job 913733, 2026-09-27.

VASP results

Supercell	Atoms	GPUs	k-points	Bands	SCF cycles	Elapsed (s)	s / SCF	E (eV/atom)	Status
3×3×3	54	1	3	136	20/20	15.3	0.76	-5.4105	converged
3×3×3	54	2	3	136	20/20	17.8	0.89	-5.4105	converged
3×3×3	54	4	3	136	20/20	18.3	0.91	-5.4105	converged
3×3×3	54	8	3	136	20/20	27.3	1.36	-5.4105	converged
4×4×4	128	1	3	321	20/20	39.6	1.98	-5.4211	converged
4×4×4	128	2	3	321	20/20	38.3	1.92	-5.4211	converged
4×4×4	128	4	3	321	20/20	36.5	1.82	-5.4211	converged
4×4×4	128	8	3	321	20/20	42.5	2.13	-5.4211	converged
5×5×5	250	1	1	626	20/20	57.6	2.88	-5.3903	converged
5×5×5	250	2	1	626	20/20	62.4	3.12	-5.3903	converged
5×5×5	250	4	1	628	20/20	75.3	3.76	-5.3903	converged
5×5×5	250	8	1	632	20/20	108.1	5.41	-5.3903	converged
6×6×6	432	1	1	1,081	20/20	113.7	5.68	-5.4111	converged
6×6×6	432	2	1	1,082	20/20	96.2	4.81	-5.4111	converged
6×6×6	432	4	1	1,084	20/20	89.3	4.46	-5.4111	converged
6×6×6	432	8	1	1,088	20/20	123.6	6.18	-5.4111	converged
10×10×10	2,000	4	1	5,004	5/5	518.4	103.69	-5.5124	not converged
12×12×12	3,456	8	1	8,648	2/2	678.6	339.29	-3.1397	not converged
14×14×14	5,488	8	1	13,728	2/2	1561.4	780.69	-3.1433	not converged
15×15×15	6,750	8	1	16,880	0/2	-	-	-	crashed
16×16×16	8,192	8	1	20,488	0/2	-	-	-	crashed

VASP GPU build, single-point SCF. 3×3×3 to 6×6×6: NELM=20, EDIFF=1E-6. 10×10×10: NELM=5, ALGO=Fast. 12×12×12 and larger: NELM=2, ALGO=Fast. SCF cycles = completed / maximum allowed (NELM). E = energy of the last SCF cycle / atom count. Converged = last energy change below the run's EDIFF. Amber rows were stopped early for timing; their E is an unconverged intermediate value, not a result. Red rows crashed before the first SCF cycle (MPI segfault, not out of memory).